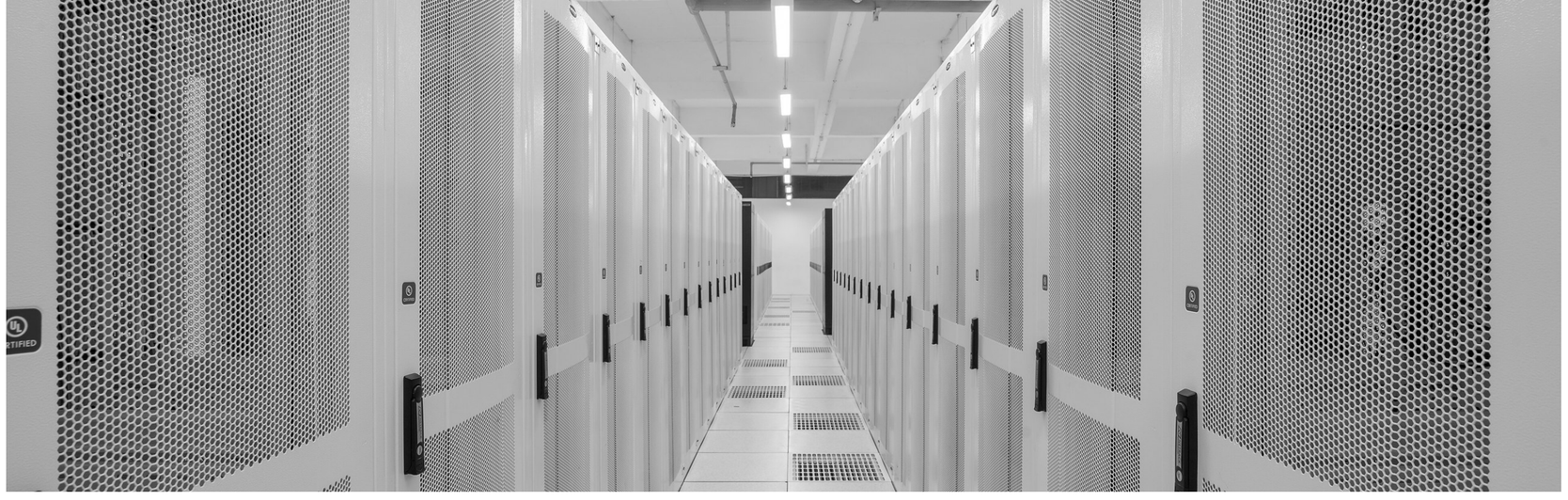


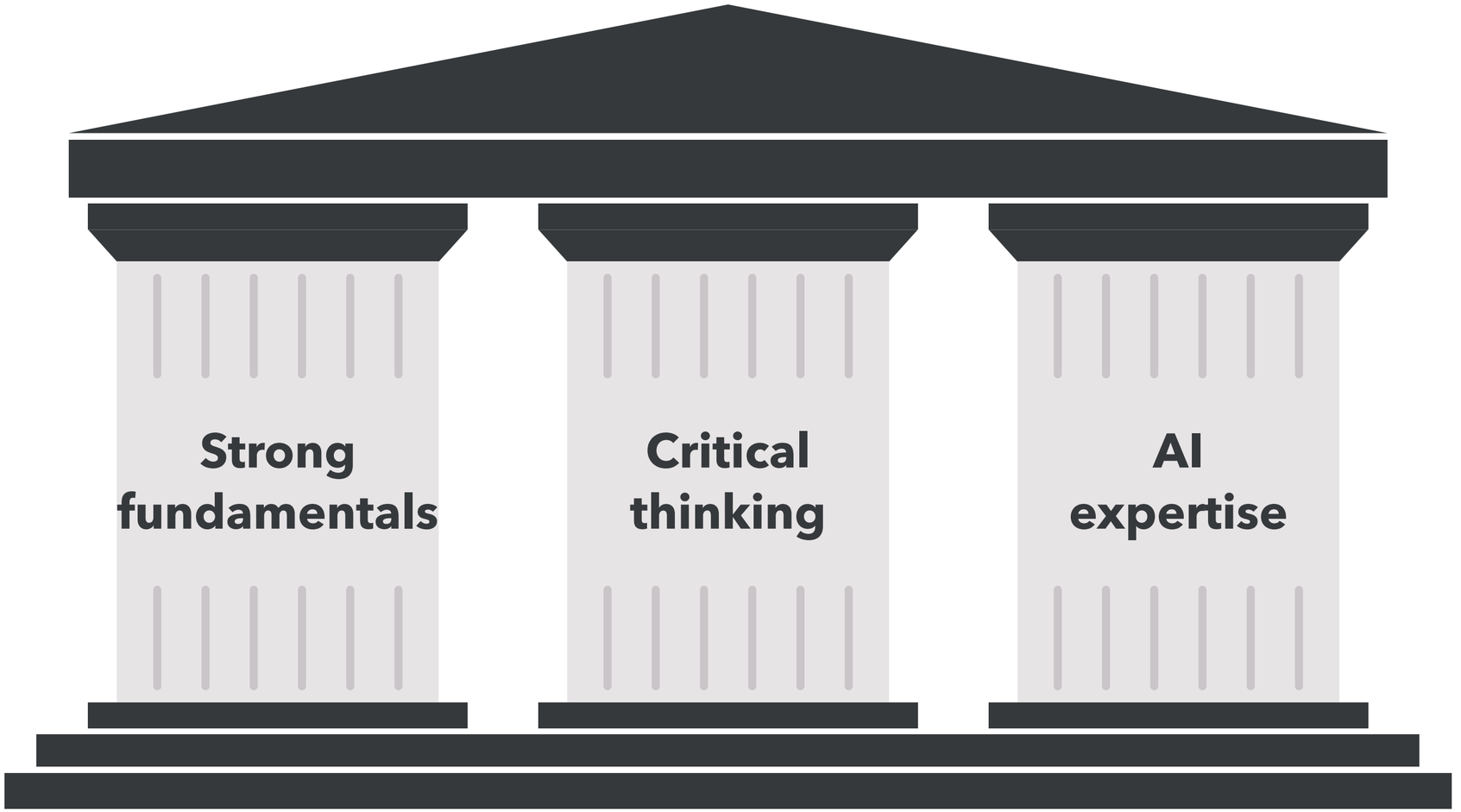
THE FINANCE CURRICULUM IN THE AGE OF AI

SESSION 1

KERRY BACK, RICE UNIVERSITY
DEPARTMENT OF FINANCE, UNIVERSITY OF ZURICH

NOVEMBER 20, 2026

PILLARS OF A FINANCE EDUCATION



KEY QUESTIONS

1. We can now go faster (and deeper). How much faster?
2. How much Excel?
3. How much Python?

MATH PRACTICE WITH GPT-4

About 1,000 high school students in Turkey. Each class session had two parts:

1. Students solved practice problems. Each classroom was randomly given plain GPT-4, a GPT-4 tutor, or no AI.
2. All students then took an exam with no AI.

Grades compared with the no-AI group	Practice problems	Exam	What students believed
Plain GPT-4	+48%	−17%	They had not learned less
GPT-4 tutor (hints, not answers)	+127%	No difference	They had done better on the exam

Bastani et al., “Generative AI without guardrails can harm learning,” PNAS, 2025

LEARNING A NEW LIBRARY WITH AI

52 software engineers who knew Python but had never used Trio, a Python library for asynchronous programming.

1. Each engineer coded two features with Trio. One group had an AI assistant. The other group coded by hand.
2. All engineers then took a quiz on Trio with no AI.

	Minutes to finish the task	Quiz score
AI assistant	23	50%
By hand	25	67%

The time difference is not significant. In the AI group, engineers who asked the AI for explanations scored 65–86%. Engineers who had the AI write or debug the code scored 24–39%.

Shen and Tamkin, "How AI impacts skill formation," Anthropic, January 2026

WRITING ESSAYS WITH CHATGPT

83% vs 11%

could not quote a sentence from the essay they had just written: ChatGPT group vs the other two groups

0 of 18

ChatGPT users quoted their own essay correctly

> 54 participants wrote essays with ChatGPT, with a search engine, or with no tools, wearing EEG headsets. The ChatGPT group had the weakest brain connectivity and the lowest sense of owning their essays. Preprint, not yet peer-reviewed.

Kosmyna et al., “Your Brain on ChatGPT,” MIT Media Lab, 2025

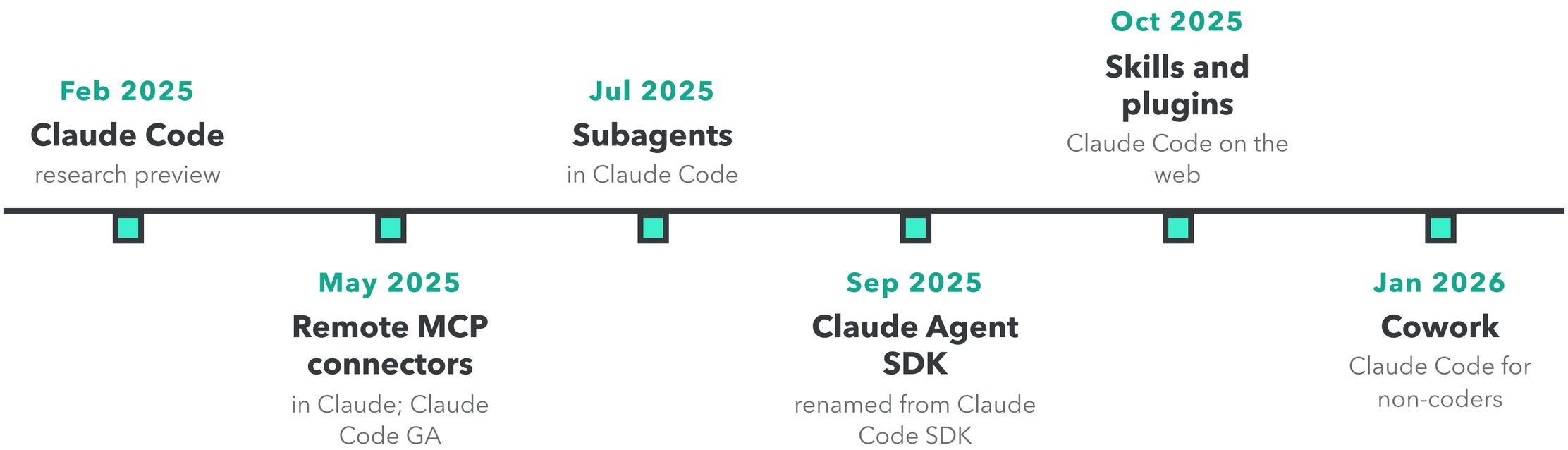
HOW MUCH EXCEL?

- Sum, average, stdev, correl, …
- The value of Excel now is to check AI's answers.
- Example: tangency.kerryback.com

HOW MUCH PYTHON?

- Types, if-else, loops, functions, objects, methods, basic pandas and numpy, what's in what library.
- Little need for Jupyter notebooks.
- Work interactively with AI instead.

TIMELINE OF AGENT TOOLS



AI AND CODING

October 2025

"They just don't work. They don't have enough intelligence … I feel like the industry is making too big of a jump and is trying to pretend like this is amazing, and it's not. It's slop."

January 2026

"I really am mostly programming in English now, a bit sheepishly telling the LLM what code to write… in words."

February–March 2026

"It is hard to communicate how much programming has changed due to AI in the last 2 months … Coding agents basically didn't work before December … I don't think I've typed like a line of code probably since December."

Andrej Karpathy: OpenAI co-founder, former Tesla AI chief, now at Anthropic

BACKTESTS FROM PLAIN ENGLISH

70–80%

of vague prompts for a backtest failed before December 2025

~80%

of such prompts, in many cases, worked the first time with Claude Opus 4.5 and GPT-5.2-Codex

> “If testing an idea used to take a month, you might say it’s not worth it. But if AI cuts that to a week or a day, suddenly there are a lot more projects you want to do. So far it hasn’t reduced headcount.”

Tal Fishman, head of fixed income quant research, Lord Abbett ($248B); AI Street, March 3, 2026

Job postings, a hiring survey, and JP Morgan

WHAT EMPLOYERS WANT

JOB POSTINGS: USING AI

Phrase	Posting	Experience
“AI tools in your work or school projects”	PIMCO, 2027 summer intern, Munich	Intern
“hands-on experience with AI-powered tools from day one”	PIMCO, 2027 summer intern, Munich	Intern
“sound judgement in technology-enabled decision making”	BlackRock, 2027 quantitative masters intern, London	Intern
“proactive about using AI where possible”	RA Capital, junior associate, Boston	About 2 yrs
“AI-forward people”	RA Capital, healthcare associate, Boston	2-5 yrs; PhD or MD
“AI tools to systematize workflows and models”	Ares, analyst, New York	1-3 yrs

Job postings, checked October 31, 2026

JOB POSTINGS: BUILDING TOOLS

Phrase	Posting	Experience
“leveraging tools like Claude Code”	D. E. Shaw, AI product analyst, Denver	None stated
“custom prompts, skill libraries, and automated workflows”	Schonfeld, AI strategy analyst, London	3-5 yrs
“we will teach the domain to the right technical candidate”	Schonfeld, AI strategy analyst, London	3-5 yrs
“It is a building role.”	Two Sigma, AI solutions developer, New York	1+ yrs; 5-10 preferred
“the judgment to figure out what to build in the first place”	Two Sigma, AI solutions developer, New York	1+ yrs; 5-10 preferred
“converting complex workflows into logical, step-by-step tasks”	T. Rowe Price, investment AI analyst, Baltimore and London	5+ or 10+ yrs

Job postings, checked October 31, 2026

JOB POSTINGS: TRANSLATING INVESTMENT PROBLEMS

Phrase	Posting	Experience
“Translate complex investment problems into structured AI workflows”	T. Rowe Price, investment AI analyst	5+ or 10+ yrs
“Translate broad investment questions into testable data science workflows”	Wellington, data science analyst, Boston	3-5+ yrs
“not just a technical interest in the data itself”	Wellington, data science analyst, Boston	3-5+ yrs
“both a hands-on technical builder and a strategic partner to investment professionals”	Carlyle, AlpInvest engineer, New York	8+ yrs

Job postings, checked October 31, 2026

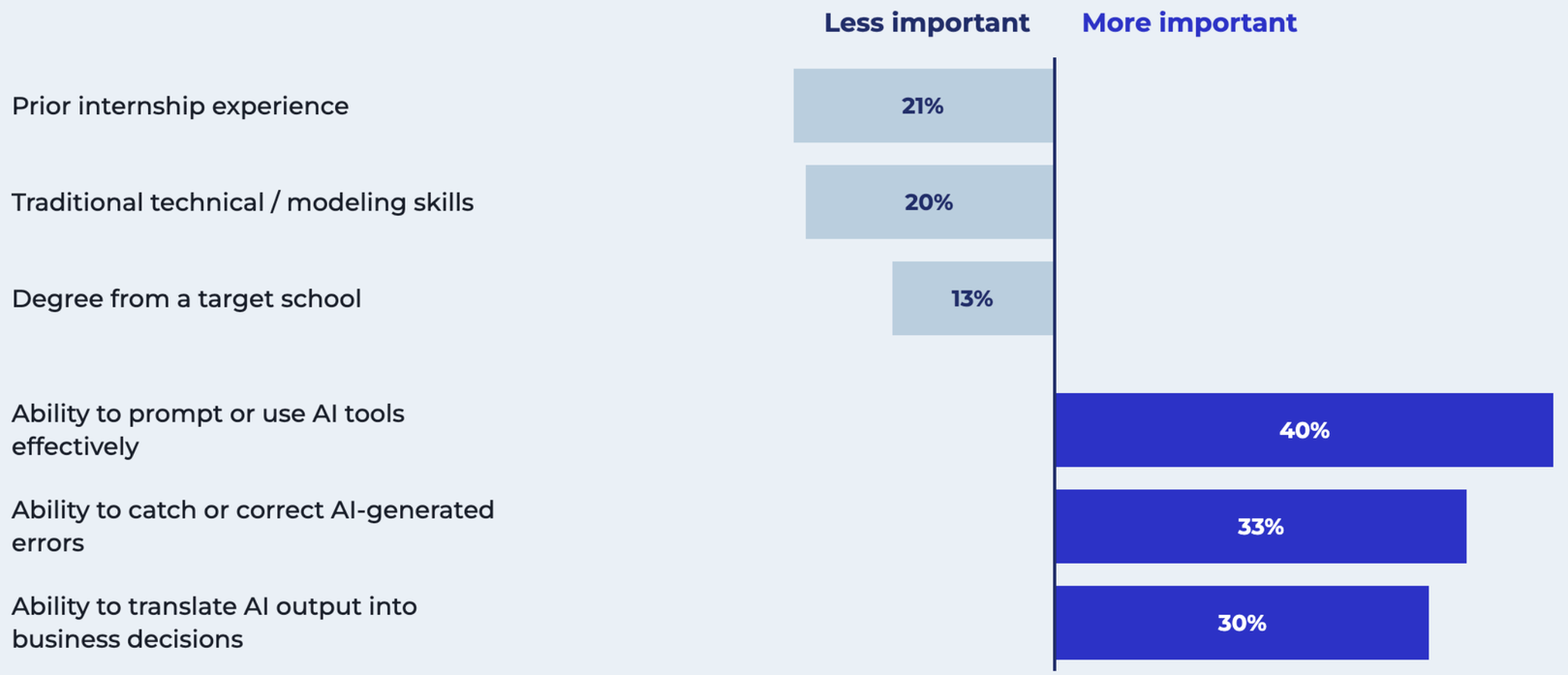
JOB POSTINGS: CHECKING THE OUTPUT

Phrase	Posting	Experience
“limitations, confidence levels, and interpretation risk”	Wellington, data science analyst, Boston	3-5+ yrs
“how these tools have evolved and where they break down”	Two Sigma, AI solutions developer, New York	1+ yrs; 5-10 preferred
“robust guardrails, compliance checks, and observability frameworks”	T. Rowe Price, investment AI analyst	5+ or 10+ yrs
“evaluation, monitoring, and human-in-the-loop patterns”	Carlyle, AlpInvest engineer, New York	8+ yrs
“appropriate audit trails”	Schonfeld, AI strategy analyst, London	3-5 yrs

Job postings, checked October 31, 2026

Is AI Fluency the New Ivy League?

The push to build AI-savvy teams has changed the industry's longstanding framework for what qualified talent looks like. Ivy League pedigree is out. AI fluency is in. Below shows the % of finance HR Professionals citing which factor will carry the most weight when evaluating candidates in their organization's next hiring cycle



Selby Jennings: The AI Fluency Gap. Research conducted by Censuswide, sample of 500 US HR Professionals in the finance industry in August, 2026.

FROM MAKERS TO CHECKERS

> Waldron said that workers would shift from being creators of reports or software updates, or “makers” in his terminology, to “checkers” or managers of AI agents doing that work.

CNBC on JP Morgan Chief Analytics Officer Derek Waldron, September 30, 2025


Critical
thinking

BANK CEOS ABOUT AI

> I need human judgment, I need human decision making, I need human accountability at the core.

HSBC CEO Georges Elhedery, Bloomberg TV interview, June 8, 2026

> The goal is to remove the repetitive, administrative work so our people can focus on what they do best—we're not replacing critical thinking and judgment, we're removing the tedious, time-consuming work that gets in the way of it.

Customers Bank CEO Sam Sidhu, statement to Fortune, April 28, 2026

WHY AI IS WEAKER ON JUDGMENT CALLS

> The gulf between LLM performance on verifiable tasks vs. unverifiable tasks that require judgment is larger than most people realize. Verifiable tasks, like coding or solving math problems, have a right answer that can be checked automatically. Judgment tasks don’t have unambiguous, ground-truth, correct answers. But they do have better and worse answers.
>
> We have much better techniques for training AI systems to do well on verifiable tasks. As a result, progress on performing such tasks has been much faster than for judgment tasks.

Andrew Ng, co-founder and former head of Google Brain, co-founder of Coursera and DeepLearning.AI · The Batch, October 16, 2026

AN EXAMPLE OF POOR REASONING

I need to wash my car. The carwash is 200 feet away. Should I walk or drive?

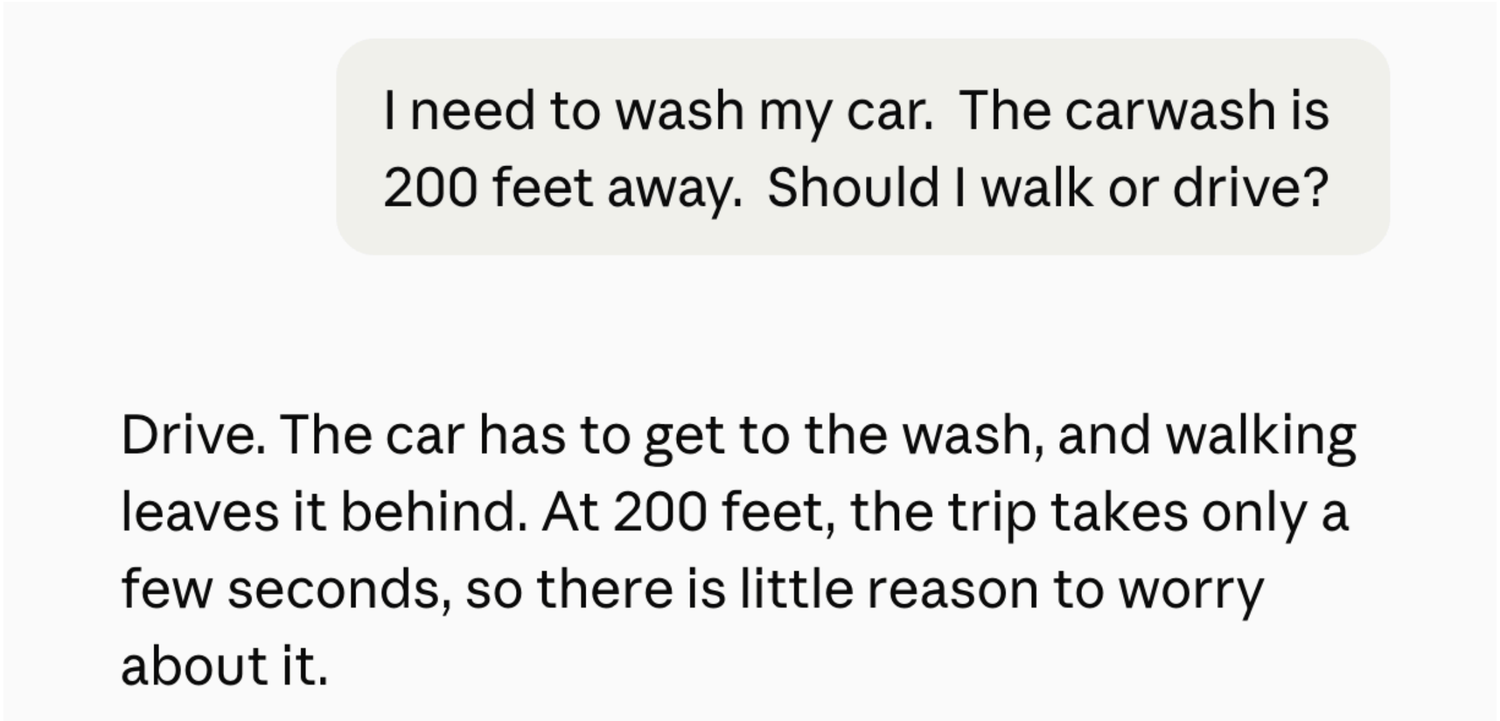
Weighing whether to walk or drive a short distance.

Walk. At 200 feet, it's only a 1-minute walk at normal pace, whereas getting the car started, driving, and parking would easily take 2–3 minutes. You'll also get a few minutes of movement in, and the car won't pick up any meaningful dirt over that distance anyway.

The only reason to drive would be if weather is genuinely unpleasant (though October in Houston is usually fine for a quick walk) or if you needed the car positioned a certain way for the wash.

Claude Haiku 4.5, an older version of Anthropic's smallest model, October 17, 2026

NEWER MODELS ARE BETTER



Claude Haiku 5.5, October 17, 2026

ANTHROPIC EXPERIMENT

1. 50 causal-inference practitioners analyze observational data on a fictional farm-advisory program, working with Claude Code (Opus 4.8), and recommend roll out or discontinue.
2. Four analytical decisions (productivity measure, identification strategy, estimand, weighting) and the final recommendation each have two defensible options.
3. For each decision, a coin flip sets which option the agent leans toward. A hidden system prompt tells the agent to hold that lean strongly and voice it.
4. The agent makes the best honest case for its lean. Participants can choose any option.

Herdağdelen, Joren and Tamkin (2026), "Who's Aligned to Whom? How AI Agents Shape Experts' Research Judgment Calls," Anthropic.

RESULTS

86%

of analytical decisions went with the agent's lean (50% by chance)

77%

repeated the agent's talking points

15%

mentioned the agent

- The four analytical leans alone predicted 82% of final recommendations, against a 61% baseline (exploratory).
- The agent's lean on the final recommendation moved it 17% of the way from discontinue to roll out (one-sided p = 0.07).
- Participants who took a side and then heard the agent lean the other way switched 60% of the time.
- More than half rated the recommendation as more their own than the agent's.

Herdağdelen, Joren and Tamkin (2026), "Who's Aligned to Whom? How AI Agents Shape Experts' Research Judgment Calls," Anthropic.

KNOWLEDGE WORKERS USING AI

Higher confidence in GenAI is associated with less critical thinking, while higher self-confidence is associated with more critical thinking.

Lee et al., survey of 319 knowledge workers on 936 tasks, Microsoft Research and Carnegie Mellon, CHI 2025 · self-reported data

WHAT DOESN'T AFFECT RELIANCE ON AI

- An AI writing assistant (GPT-3.5 and GPT-4) offered autocomplete suggestions with a set position on societal issues.
- 2,582 participants. Their attitudes shifted toward the assistant's position.
- Warning participants about the bias, before or after writing, did not reduce the shift.

Williams-Ceci et al. (2026), "Biased AI Writing Assistants Shift Users' Attitudes on Societal Issues," Science Advances.

WHAT DOES AFFECT RELIANCE ON AI

What the LLM did	Study	Finding
Stated uncertainty in the first person (“I’m not sure, but…”)	Kim et al. (2024), 404 participants, medical questions	Agreed with the LLM less and answered more accurately
Cited sources	Kim et al. (2025), 308 participants	Relied less on incorrect answers
Explained its answer	Kim et al. (2025), 308 participants	Relied more on both correct and incorrect answers
Added a disclaimer, highlighted uncertain words, or implied the answer without stating it	Bo, Wan and Anderson (2025), 400 participants	Relied less on wrong answers, but also less on right ones

All three studies use questions with a correct answer.

A POSSIBLE CLAUDE.MD SECTION

```
## Judgment calls

A judgment call is a decision with more than one defensible answer:
a method, an assumption, an interpretation, or a recommendation.

1. When you reach a judgment call, run each option and check
   whether the result changes.
2. If the result changes, stop and name the judgment call.
   Make the best case for each option, with equal effort.
   Refuse to do more than that. Do not recommend a choice.
3. If the result does not change, list the judgment call
   at the end of your answer.
```

PROMPTING

Say what you want

Not what to do.

Ask for the best way (plan) first

Then tell it to proceed.

Ask AI to interrogate you

Let it decide what it needs to know.

Beware of tunnel vision

Ask: is there a better path?

Steelman

“What is the strongest argument against this?”

Numbers need code

“Compute this in Python.”

TEACHING STUDENTS TO CHECK AI WORK

Have AI explain its work

"Explain what you did step by step. Did you make any decisions that we have not discussed?"

Use another AI agent to check the work

A fresh conversation or a subagent (a second agent that the first agent starts), given your goals and the first agent's explanation.

Have AI build a spreadsheet when possible

Then check the formulas. Every number traces back to its inputs.

Run the same code on small examples

Ask AI to run its code on a few rows you can check by hand.

Most AI coding errors arise from misunderstandings and unstated assumptions — and from errors that people also make, for example, merging and filtering in the wrong order.

SOFTWARE

SANDBOXED AGENTS

What they are

- Cowork (and Codex, …) runs in a virtual machine or container. The container has a whitelist of web addresses it can visit, including Anthropic, GitHub, Python, and a few others.
- It can do web fetch and web search because they are run at Anthropic. It only needs to reach Anthropic to run them.
- You can edit the whitelist (allowlist) in Cowork (but not Codex, I think) but it is reported to be buggy.

When the sandbox binds

- Cannot reach URLs from code.
- Cannot use CLIs of external services or run curl commands to use APIs.
- Cannot run grep and glob to search files outside the project folder.

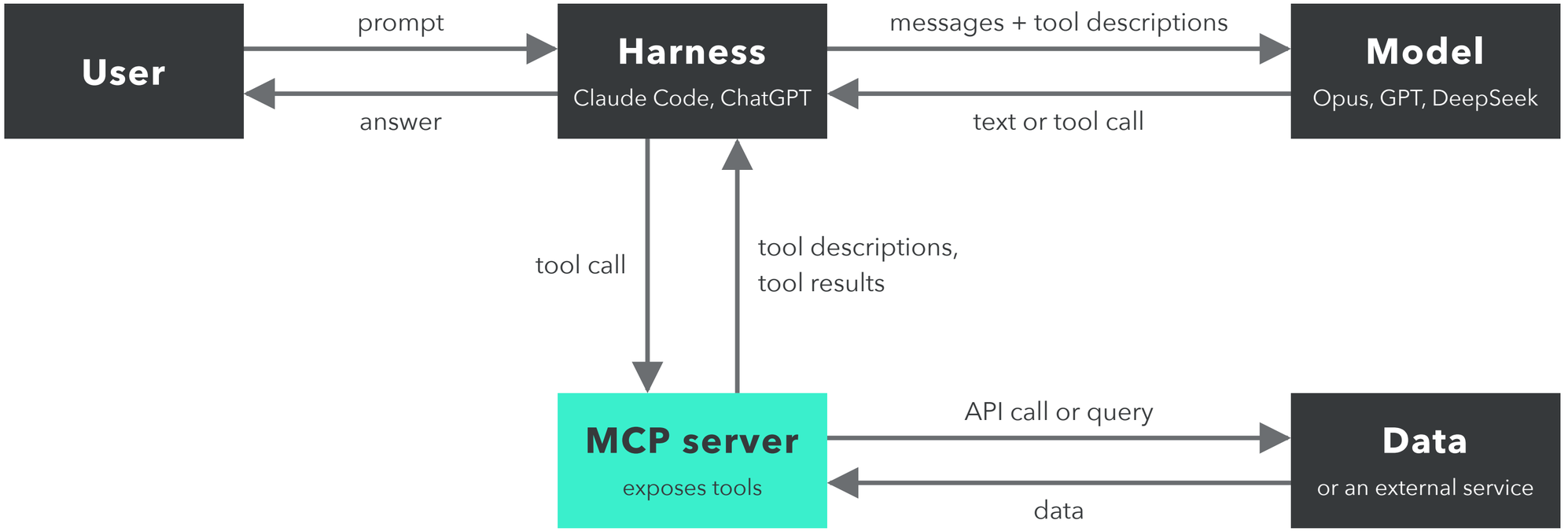
NON-SANDBOXED VERSIONS OF CLAUDE

- Code option in Claude Desktop
- Claude Code CLI
- Claude Code extension for VS Code
- Academic Studio, my fork of VS Code that integrates Claude for non-coders: academic-studio.com

MCP servers connect students’ agents to data

DATA

WHAT AN MCP SERVER IS



WHAT “EXPOSES TOOLS” MEANS

- The server gives the harness a text description of each tool: name, purpose, and inputs.
- The harness sends the descriptions to the model with each request.
- The model returns a tool call. The harness sends it to the server, and the server runs the tool.

WHAT A SERVER SENDS FOR ONE TOOL

```
{
  "name": "french_data",
  "description": "One table of a Ken French dataset as CSV. Returns are in decimals.",
  "inputSchema": {
    "type": "object",
    "properties": {
      "name":  {"type": "string"},
      "table": {"type": "integer", "default": 0}
    },
    "required": ["name"]
  }
}
```

> The model sees this text, not the function's code.

Simplified from the server's response to a `tools/list` request.

END-OF-DAY PRICES

Server	Free coverage	Additional paid coverage	Hosted or local
Shibui	US (NYSE, NASDAQ), 64 years; no key; fair-use limits	None listed	Hosted
Equibles	US daily bars; 100 requests/day	10,000 or 100,000 requests/day	Hosted; local
MaverickMCP	Global, through Yahoo; no key	None	Local
EODHD	20 calls/day; limited data	All exchanges, $19.99/mo	Hosted
Alpha Vantage	25 requests/day	75 requests/min, $49.99/mo	Hosted; local
FMP	250 calls/day; end-of-day only	Global by plan, from $29/mo	Hosted (sources disagree)
Massive	US; 5 calls/min; 2 years	20+ years, $199/mo	Local; needs key

Vendor pages and comparison pages, November 1, 2026. Not tested.

INTRADAY AND REAL-TIME PRICES

Server	Free coverage	Additional paid coverage	Hosted or local
tradingview-mcp	Live data and technicals; no key	Hosted, $9 or $29/mo	Local; hosted is paid
Mcprice	Live stocks and ETFs (Yahoo), crypto (Binance); no key	None listed	Not confirmed
Alpha Vantage	Delayed; 25 requests/day	15-minute delay from $49.99/mo; real time from $99.99/mo	Hosted; local
EODHD	Not in the free plan as far as listed	End-of-day plus intraday, $29.99/mo	Hosted
Equibles	None	Quotes delayed 15 minutes on Plus	Hosted only
Massive	None; free plan is end-of-day	Real time and ticks from $199/mo	Local; needs key

Vendor pages and comparison pages, November 1, 2026. Not tested.

STATEMENTS AND FUNDAMENTALS

Server	Free coverage	Additional paid coverage	Hosted or local
Shibui	US statements, 35+ years; daily valuations; no key	None listed	Hosted
Equibles	US fundamentals; 100 requests/day	Higher limits; multiples and screener are hosted only	Hosted; local
FMP	Statements and ratios, 30+ years on major names; 250 calls/day	Global by plan, from $29/mo	Hosted (sources disagree)
EODHD	Limited data; 20 calls/day	Fundamentals, $59.99/mo	Hosted
Alpha Vantage	Basic fundamentals; 25 requests/day	Higher limits, $49.99/mo	Hosted; local
sec-edgar-mcp	XBRL statements with SEC links; no key	None	Local
edgar-mcp	XBRL highlights: revenue, net income, assets, EPS; no key	None	Local
MaverickMCP	Fundamentals through Yahoo; no key	None	Local

Vendor pages and comparison pages, November 1, 2026. Not tested.

SEC FILINGS AND EARNINGS CALLS

Server	Free coverage	Additional paid coverage	Hosted or local
Shibui	6.4 million filing records, 10,000+ companies; no key	None listed	Hosted
Equibles	Filings; 100 requests/day	Higher limits; earnings calls and guidance are hosted only	Hosted; local
edgar-mcp	10-K, 10-Q, 8-K; full-text search of about 10 years; no key	None	Local
sec-edgar-mcp	Filings with SEC links; no key	None	Local
FMP	SEC filings; 250 calls/day	From $29/mo	Hosted (sources disagree)

Vendor pages and comparison pages, November 1, 2026. Not tested.

INSIDER TRADES AND HOLDINGS

Server	Free coverage	Additional paid coverage	Hosted or local
Equibles	13F, insider, congressional, and short data; 100 requests/day	10,000 or 100,000 requests/day	Hosted; local
sec-edgar-mcp	Forms 3, 4, and 5; no key	None	Local
Finnhub	Insider transactions, earnings surprises, news sentiment; about 60 calls/min	Price not confirmed	Sources disagree

Vendor pages and comparison pages, November 1, 2026. Not tested.

OPTIONS

Server	Free coverage	Additional paid coverage	Hosted or local
triplom	Chains with Greeks, through Yahoo; no key for this data	None	Local; setup asks for a FRED key
Equibles	None	Delayed chains with implied volatility and Greeks on Plus	Hosted only
Massive	None	Options prices without Greeks; price not confirmed	Local; needs key
Alpha Vantage	Not confirmed	Options; plan not confirmed	Hosted; local
EODHD	Not confirmed	End-of-day options; plan not confirmed	Hosted

Vendor pages and comparison pages, November 1, 2026. Not tested.

FACTORS, MACRO, AND ENERGY

Server	Free coverage	Additional paid coverage	Hosted or local
Equibles	FRED, CFTC, and CBOE data; 100 requests/day	10,000 or 100,000 requests/day	Hosted; local
Alpha Vantage	Macro series; 25 requests/day	Higher limits, $49.99/mo	Hosted; local
triplom	FRED, with a free FRED key	None	Local
EIA server	All of EIA's API v2; no key	None	Hosted; local

Vendor pages and comparison pages, November 1, 2026. Not tested.

EUROPEAN EQUITIES

Server	Free coverage	Additional paid coverage	Hosted or local
tradingview-mcp	Prices, screens, and technicals: Euronext, LSE, Xetra, Milan, SIX, Madrid	Hosted, $9 or $29/mo	Local; hosted is paid
EODHD	20 calls/day; limited data	Prices $19.99/mo; fundamentals $59.99/mo	Hosted
Equibles	Director trades, fund holdings, and short positions; 100 requests/day	10,000 or 100,000 requests/day	Hosted
esef-filings	Annual IFRS facts from filings.xbrl.org; no key	None	Not confirmed
eu-filings	ESEF filings for FR, DK, GB, LT, UA; GLEIF; no key	None	Local
Finance Database	300,000+ symbols; no prices or statements; no key	None	Hosted; local
RegistrumUK	UK Companies House; small daily caps, then a free key	Volume plans; price not confirmed	Hosted

Vendor pages and comparison pages, November 1, 2026. Not tested.

CREATING AN MCP SERVER

- AI can write MCP servers for us.
- Each tool is a function. It can call a web API or a Python library: yfinance for Yahoo Finance and pandas-datareader for French's data library.

```
@mcp.tool()
def french_data(name: str, table: int = 0) -> str:
    """One table of a Ken French dataset as CSV. Returns are in decimals."""
    data = web.DataReader(name, "famafrench", start="1900-01-01")
    return (data[table] / 100).to_csv()
```

The docstring and type hints become the tool description on the earlier slide.

HTML pages and workbooks

DELIVERABLES

AI OUTPUTS

- We should emphasize AI outputs.
- HTML is the most useful: easy to read, easy for AI to write, and interactive.
- Excel is a fallback to check the AI output.

NEW STUDENT DELIVERABLES

- Software apps: every course
- Skills: every course
- Custom agents: AI course

TWO-STAGE DCF: HTML PAGE

John B. Sanfilippo & Son, Inc. (JBSS)

Two-stage DCF. Fiscal 2026, ended June 25, 2026, is year 0. Figures are in USD thousands, except per-share value. Change any input, and everything below recalculates. The same model, with live formulas, is in the workbook: jbss.xlsx.

Value per share	Equity value	Enterprise value	Terminal value share of enterprise value
$120.45	1,418,569	1,517,823	77.7%

Reset to baseline

Assumptions

Assumption	Year 1	Year 2	Year 3	Year 4	Year 5	Terminal
Growth and margin						
Sales growth	4.5 %	6.5 %	6 %	4.5 %	3.5 %	3.5 %
EBITDA margin	10.2 %	10.6 %	10.8 %	10.8 %	10.8 %	10.8 %
Property, plant and equipment						
Next-year sales / net PP&E	5.25 ×	5.6 ×	5.9 ×	6.15 ×	6.3 ×	6.3 ×

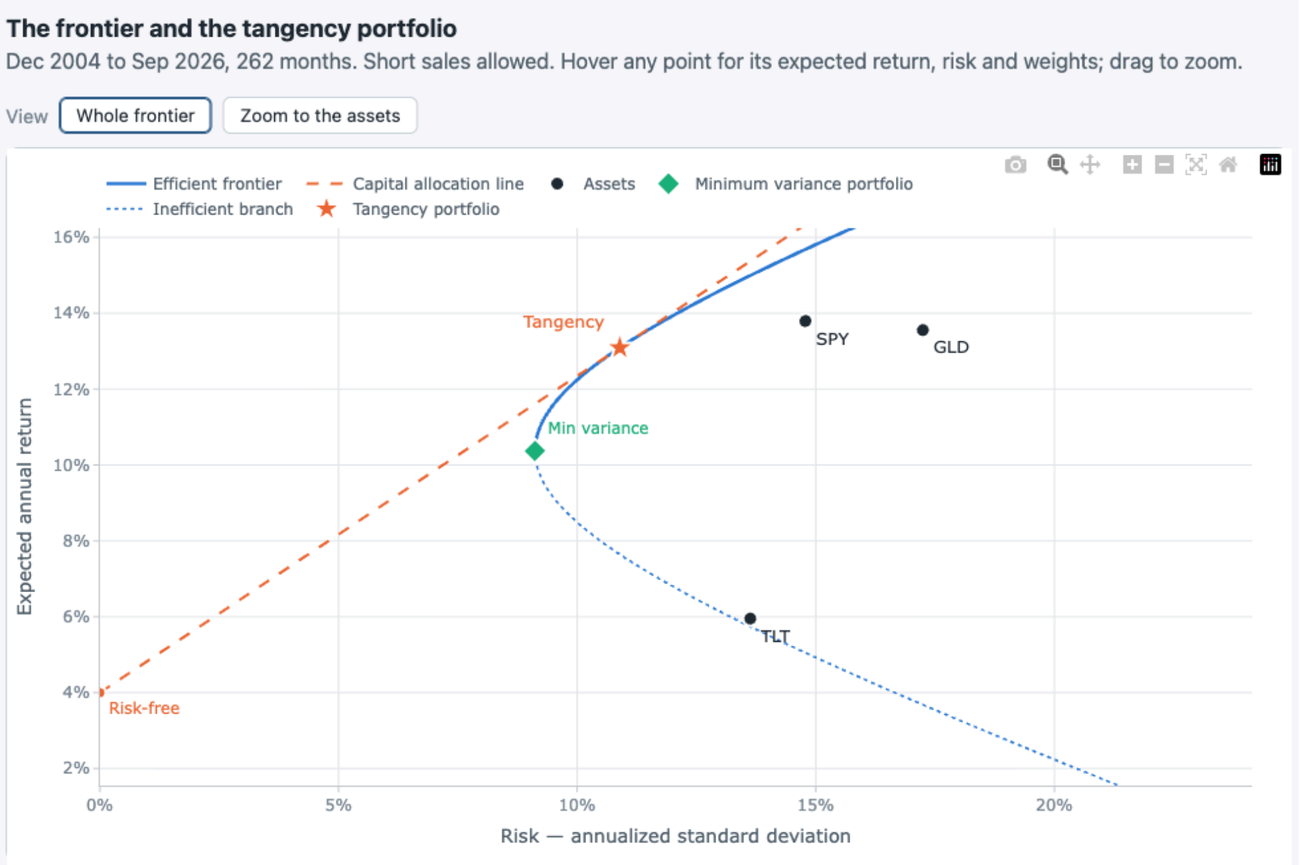
Open the page. John B. Sanfilippo & Son, fiscal 2026 as year 0. Every input is editable.

TWO-STAGE DCF: WORKBOOK

Sheet	Contents
Inputs	The same inputs as the HTML page
Model	The forecast; every cell is a formula over Inputs
Bridge	Enterprise value to value per share
Sensitivity	Value per share by WACC and terminal growth; a snapshot
Historical	Historical driver ratios

Open the workbook. 1,447 formula cells. Value per share is $120.45 on the page and in the workbook.

TANGENCY PORTFOLIO: APP



tangency.kerryback.com. SPY, TLT, and GLD, December 2004 to September 2026. The page also gives the weights and a workbook to download.

TANGENCY PORTFOLIO: WORKBOOK

1. Means
2. Standard deviations
3. Correlations
4. Covariance matrix
5. Inverse covariance matrix
6. Inverse covariance matrix times risk premia
7. Scaled to the tangency portfolio

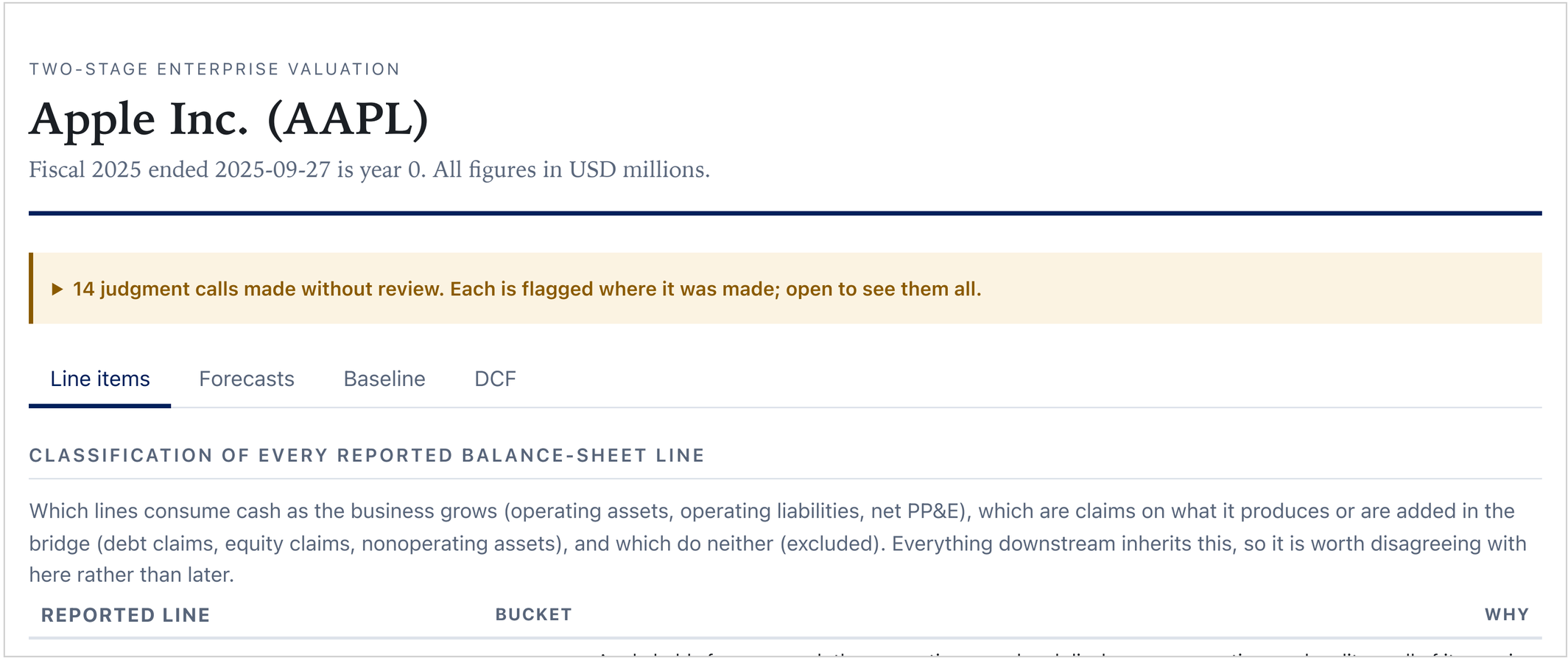
Open the workbook. SPY, TLT, and GLD, December 2004 to September 2026.

CRITICAL THINKING AGAIN

AUTOMATE THE MECHANICAL STUFF

- Grab data: numerical and text.
- Display relevant parts and tag judgments to be made.
- Students can build these.
- Spend class time on judgment parts.

VALUATION: SKILL



Open the page. Apple, fiscal 2025 as year 0. Line items, Forecasts, Baseline, and DCF tabs, with 14 judgment calls flagged.


Strong
fundamentals

FLIPPING THE CLASSROOM

- We have to expect out-of-class assignments will be done by AI.
- Assessing in class by cold-calling remains viable.
- This suggests flipping the classroom: providing lectures beforehand and spending class time interacting (case-style teaching).
- Ask students to share screens and defend their work and assumptions.

MINI CASES

1. Use AI to write cases.
2. Give students 5 minutes to read.
3. Discuss on whiteboard.
4. Give students 10 minutes to do analysis with AI.
5. Call on students to present and explain.

AI TUTORS

- AI is a great teaching tool.
- We can add value by controlling what it emphasizes and how it explains things.
- Customized AI tutors will replace textbooks.
- Examples: tutors.kerryback.com

AI TUTOR VS ACTIVE-LEARNING CLASS

2×

median learning gain, AI tutor vs active-learning class

49 vs 60

minutes on task

> 194 students in Harvard introductory physics. Each student learned one topic from the GPT-4 tutor at home and another in class. The tutor was prompted with the instructor's step-by-step solutions and guided students through each part of each problem.

Kestin et al., "AI tutoring outperforms in-class active learning," Scientific Reports, 2025

Man Group, HRT, Balyasny, a live test, and Alpha Arena

TRADING AGENTS

TWO WAYS FUNDS USE AI FOR TRADING

Frontier models as researchers

Claude, GPT, and Gemini propose ideas, write code, read filings, and run backtests. The signals are conventional.

Man Group, Lord Abbett, Balyasny

Models trained on market data

A transformer is trained on order-book events instead of text.

Hudson River Trading

AI Street, 2025–2026

MAN GROUP: ALPHAGPT

Idea generator
Proposes hypotheses at scale

→

Code implementer
Writes Python against Man's research stack and runs the backtest

→

Evaluator
Checks significance, risk, and economic sense

- Output then goes to the same human review and investment committee as human research.
- Built on frontier models. Man does not train its own model.
- Problems: hallucination, look-ahead bias, multiple testing, token costs.

Man Numeric (Man Group, $214B); AI Street, November 20 and December 18, 2025

ALPHAGPT: RULES AGAINST DATA MINING

“Every idea must be hypothesis-driven, with a clear economic rationale. Once you start backtesting, you can’t change the hypothesis without risking data mining. AI follows the same rules.”

Ziang Fang, senior portfolio manager, Man Numeric; AI Street, December 18, 2025

HUDSON RIVER TRADING

100+ TB
of equities, futures, and crypto order data

20+ yrs
of history, “trillions of tokens”

~10%
of US equity trading volume

- The sequence is market events: order placements, cancellations, fills.
- Each position in the sequence covers a fixed time window or a fixed number of events. HRT’s token format is not public.
- “As I increase the model size, the model continues to improve.”

Marc Khoury, HRT AI team, conference talk; AI Street, January 15, 2026

LOBERT: ONE TOKEN PER MESSAGE

Field	Values in the token
Side	buy, sell
Message type	new order, edit, delete, execution, hidden order
Price: ticks from the best opposing quote	0, 1, 2, 3, 5, 10
Volume	0, 50, 100, 200; exactly on the level or between

- Example token: new sell order, 3 ticks away, volume 100, exact. The training data produces 293 distinct tokens.
- The exact price distance, exact volume, and time since the previous message enter as continuous values beside the token.
- The model predicts the next token as a classification over the 293 tokens, and the exact values with three regression heads.
- Trained on 470 million Nasdaq messages (AAPL, INTC, MSFT, FB; May to September 2015) with 1.1 million parameters. The full next message is predicted correctly 27.8% of the time, against 6.1% for an earlier model.

Linna, Baltakys, Iosifidis, and Kanniainen, Tampere University, "LOBERT: Generative AI Foundation Model for Limit Order Book Messages," arXiv 2511.12563, 2026; AI Street, January 2, 2026

BALYASNY: MERGER ARBITRAGE

1,648
deals over $1B in 42 countries, January 2022 to December 2025

404
held-out test deals, February to December 2025

12
research agents per deal

Stage	Agents
Setup	Ticker resolution; deal card (terms, dates, fees, voting thresholds)
Research	Filings; ownership; market view; climate; regulatory risk; financing risk; precedents; timeline
Check	Gap analysis
Forward look	Catalyst tracker

Balyasny Asset Management ($38B); AI Street, September 16, 2026; Jajal et al., “Global Merger-Arbitrage Forecasting with Language Models,” Balyasny, ICML 2026

BALYASNY: THE MARKET BENCHMARK

The target's price is a mix of the deal value S^+ and the fallback value S^-:

$$S_t = p_t S_t^+ + (1 - p_t) S_t^- \quad \Longrightarrow \quad p_t = \frac{S_t - S_t^-}{S_t^+ - S_t^-}$$

- S^+ for a cash deal: the offer, discounted to the expected close (the US median of 175 days if the company gives no guidance).
- S^-: the 20-day average price before the announcement, moved by beta times the market return since then.
- p_t is then Platt-scaled on the training data.

Jajal et al., "Global Merger-Arbitrage Forecasting with Language Models," Balyasny, ICML 2026

BALYASNY: TRAINING THE MODEL

1. An agent writes a post-mortem of each training deal. Only this agent may use the open web.
2. GPT-5.1 writes a report as of the forecast date, citing only information public on that date.
3. Merger-arb specialists review the agents' work on deals they covered.
4. Target probabilities blend the market-implied probability with the realized outcome. The market's weight starts at 0.3 and falls to 0 as the deal ends. Training on outcomes alone makes early forecasts overconfident.
5. GPT-4o is fine-tuned on the reports and probabilities.

> 84% of deals closed as announced, 2.7% ended with a higher bid, and 13.7% ended with no higher bid. Terminated deals are oversampled in training.

Jajal et al., "Global Merger-Arbitrage Forecasting with Language Models," Balyasny, ICML 2026

BALYASNY: RESULTS

Model	Brier score	Calibration error	Correlation with market
Market-implied, Platt-scaled	0.199	0.016	1.00
XGBoost: market probability and 26 deal features	0.186	0.035	0.75
Claude 3.7 Sonnet	0.259	0.146	0.29
Gemini 3 Pro	0.231	0.118	0.34
GPT-5.1	0.214	0.110	0.36
GPT-4o	0.201	0.089	0.30
GPT-4o with isotonic calibration	0.186	0.075	0.31
GPT-4o fine-tuned	0.151	0.039	0.36

Test set: 404 deals, February to December 2025. Class-balanced Brier score for a positive outcome for target shareholders; lower is better. All language models get the same agent research and have knowledge cutoffs before the test period.

Jajal et al., "Global Merger-Arbitrage Forecasting with Language Models," Balyasny, ICML 2026

BALYASNY: WHAT EACH PART ADDS

Removed from the system	Brier score
Nothing	0.151
Hindsight-guided reports	0.183
Market-implied probability in the targets	0.181
Half of the training deals	0.177
All agents except the deal card	0.172
Oversampling of terminated deals	0.170

Jajal et al., “Global Merger-Arbitrage Forecasting with Language Models,” Balyasny, ICML 2026

BALYASNY: THE 100 WORST FORECASTS

60
missing, stale, or wrong context

20
signal in the prompt, under-weighted

17
miscalibration

0
hallucination

Forecast	Outcome	Missing from the context
0.98	Failed	Tender-offer terms: two-thirds minimum acceptance, offer price
0.81	Failed	A blocking shareholder's stake of more than 30%
0.65	Failed	ISS and Glass Lewis recommendations against the deal
0.07	Completed	The formal merger announcement on the forecast date

Causes classified by GPT-5. The authors turned off open-web search to prevent look-ahead bias. Jajal et al., "Global Merger-Arbitrage Forecasting with Language Models," Balyasny, ICML 2026

A LIVE TEST

~50%

top-20 portfolio, nine months

~26%

Russell 1000, same period

18.5 → 1.6

basis points of daily alpha, trading one day later

- Each night, April 2025 to January 2026, a frontier model with web search scored every Russell 1000 stock from −5 to +5.
- No news or filings were given to the model.
- Top-scored stocks outperformed. Bottom-scored stocks did not reliably underperform.

Chen and Pu, Peking University, “Autonomous Market Intelligence,” January 2026; AI Street, February 19, 2026

ALPHA ARENA

6 of 32

model results finished with a profit. Eight models, including Claude, Gemini, ChatGPT, and Grok, traded US tech stocks with $10,000 each in four competitions. The combined portfolio lost about a third of its capital.

Nof1; Bloomberg; AI Street, May 7, 2026

LOOK-AHEAD BIAS

Case	Protection
Man Group	Hypothesis fixed before the backtest
Balyasny	Test set split by announcement date; model knowledge cutoffs before the test period; no open-web retrieval
Peking University	Live test, after the model's training data
Alpha Arena	Live trading
JPMorgan regime agent	Backtest only, 2006–2026; the researchers state the caveat

AI Street, 2025–2026

Changing the weights, with finance examples

FINE-TUNING

THREE WAYS TO CHANGE WHAT A MODEL DOES

	What changes	Use when
Prompting	the text sent with the request	the model can do the job and needs telling
Retrieval	documents put into the context	it needs facts it was not trained on
Fine-tuning	the weights	the behavior has to change: a format, a style, a judgment call

> Fine-tuning does not add knowledge reliably.

WHEN FINE-TUNING IS WORTH IT

A fixed output format

A label, a JSON record, a field in another system. Prompting gets the format most of the time.

Many examples, one narrow job

Thousands of labeled cases already exist, and the job will run millions of times.

Cost per call

A small fine-tuned model can match a large prompted one on one narrow task, at a fraction of the cost per call.

FINE-TUNING IN FINANCE

Firm	What was fine-tuned	Reported result
Bridgewater, with Thinking Machines	An open-weight model on examples labeled by its investment experts: is a document relevant, and where does the boilerplate begin	84.7% average accuracy on six tasks; frontier models with expert prompts in the mid-to-high 70s; 13.8× cheaper per task
RavenPack	Nemotron 3 Nano (4B) on labels from Nemotron 3 Super (120B)	Matched the large model on about 96% of sentence labels at 39× the throughput
Balyasny	GPT-4o on hindsight-guided merger-arbitrage reports	Brier score 0.151, against 0.199 for market-implied odds
JPMorgan (patent application)	An LLM with LoRA, labeled by forward-return quintiles relative to peers	Buy, hold, and sell ratings; no results reported
Mantic	gpt-oss-120b, by reinforcement learning on about 10,000 yes-or-no questions	A published research experiment; Mantic sells forecasts to hedge funds

AI Street: July 2, May 19, September 16, September 24, and October 22, 2026; Jajal et al., ICML 2026

BRIDGEWATER: AN EXPERT-LABELED MODEL

84.7%

average accuracy of the fine-tuned model on six relevance tasks

mid-to-high 70s

GPT, Claude, and Gemini with expert prompts

13.8×

lower cost per task

- Tasks: is a financial article, central bank document, or research report relevant, and where does the useful part of a document or email end and the boilerplate begin.
- When the model disagreed with a label, the example went back to Bridgewater's experts.

Bridgewater and Thinking Machines; AI Street, July 2, 2026

FINE-TUNE, OR NOT?

> "A fine-tune can be justified anywhere there is a proprietary data moat, a task flow that appears fundamentally different, or a reliance on specific properties of your company that wouldn't have been represented in the training data."

Lucas Baker, head of LLM R&D, Jump Trading; AI Street, September 22, 2026

> "It's built at the application layer. We haven't hit the limits of frontier models yet. Some tasks may eventually require fine-tuning or custom models, but the base models keep improving."

Ziang Fang, Man Numeric, on AlphaGPT; AI Street, December 18, 2025

DISTILLATION

1. Run a frontier model on your inputs and keep its answers.
2. Treat those answers as labels.
3. Fine-tune a small open-weights model on them.

> RavenPack did this with Nvidia's Nemotron models. Read a provider's terms before doing it with a commercial model's output.

FULL FINE-TUNING AND LORA

Full fine-tuning

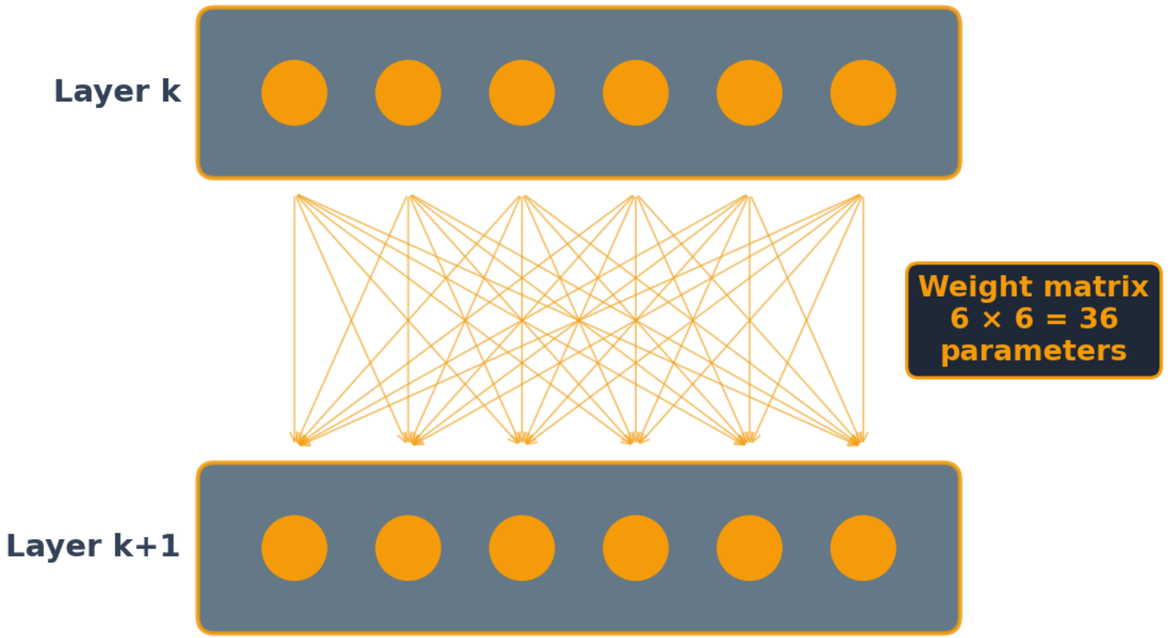
Update every parameter. Needs GPU memory for the weights, their gradients and the optimizer state, and produces a complete copy of the model.

LoRA

Freeze the original weights. Train two small matrices per layer instead. In the notebook later in this section, that is under 0.1% of the model's parameters.

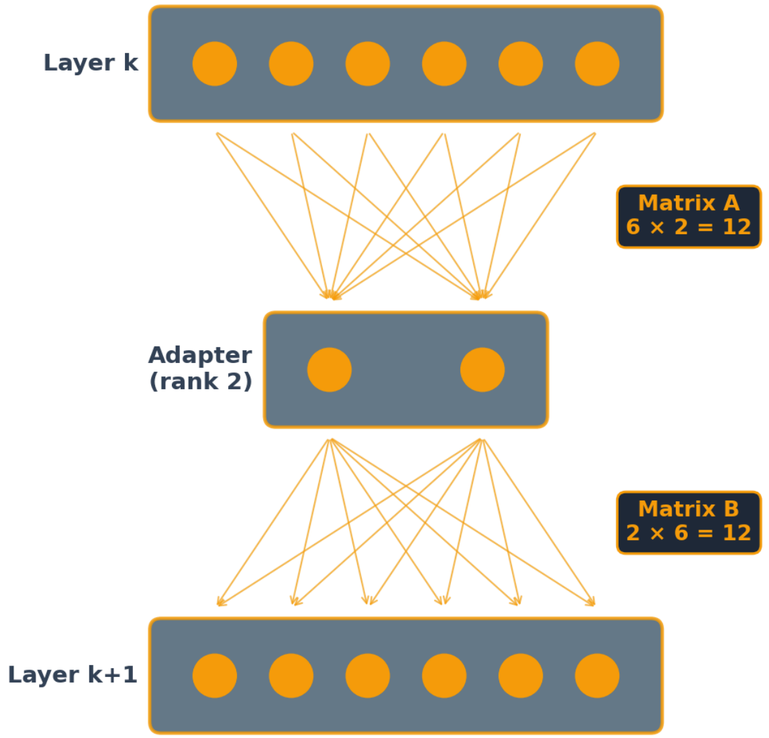
Hu et al., 2021: low-rank adaptation.

THE WEIGHT MATRIX



Six neurons feeding six neurons: 36 weights, one per connection. Full fine-tuning retrains all of them.

THE LOW-RANK PATH



Total: 24 trainable parameters (vs. 36 for full fine-tuning)

The 36 frozen connections are not drawn. The trained path runs through two neurons: rank 2.

LORA: THE ARITHMETIC

A compresses the input down to the rank and B expands it back up, so the product BA has the same shape as the original weight matrix W and can be added to it:

$$W' = W + BA$$

> There is no activation function in the adapter path, which is why BA is a matrix and the addition is possible. Once merged, the model has the architecture and the speed it had before.

AT REAL SCALE

The six-by-six picture is a diagram. A transformer's layers are 4096 × 4096 and larger.

16.8M

parameters in one 4096 × 4096 layer

131K

trained by a rank-16 adapter on it

128×

fewer parameters to train

> The adapter is also what you store and ship: a file of megabytes, where full fine-tuning leaves you a second copy of the whole model.

QLORA

The base model, quantized

Weights stored at 4 bits instead of 16. Gemma 3 1B goes from about 2 GB to about 800 MB.

Adapters on top of it

The frozen weights are read at 4 bits; the small matrices train at full precision. A 70-billion-parameter model fits on one 48 GB GPU.

Dettmers et al., 2023. This is what puts fine-tuning on a free Colab T4, which has 16 GB.

HANDS-ON: FINE-TUNE GEMMA 3 1B

Google Colab · free T4 GPU · about 15 minutes of training

1. Load Gemma 3 1B at 4 bits.
2. Attach rank-16 adapters to the attention layers, `q_proj` and `v_proj` only.
3. Train on Financial PhraseBank: 2,264 news sentences that every annotator labeled the same way, positive, negative or neutral.
4. Run three test sentences through the model before and after.

Open the notebook in Colab

> Needs a Hugging Face account, a read token, and Google's Gemma license accepted at huggingface.co/google/gemma-3-1b-it.

EXERCISE: FINE-TUNE GEMMA

1. Run the notebook as far as the baseline cell, and read what the base model says about the three test sentences.
2. Train, then compare the answers before and after.
3. The base model is about 2 GB and the saved adapter is 14 MB. Say what has to be on the serving machine for the fine-tuned model to answer, and what changes if you merge the adapter into the weights instead.